



MSML610: Advanced Machine Learning

Lesson 11.1: Decision-Making with Causal Models

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References:

- *Why Prediction Is Not Enough*
 - Prediction Pipelines vs. Decision Pipelines
 - When Prediction Fails
 - Causal Models as the Foundation for Decisions
- Foundations of Decision Theory
- Decision Support with Causal Models
- Sequential Decision-Making and Active Learning
- Uncertainty in Causal Decisions

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From Predictions to Decisions

- **Prediction** answers: *“What will happen?”*
 - Input: features X
 - Output: estimate $\hat{y} = \mathbb{E}[Y|X]$
 - Optimized for accuracy on a fixed distribution
- **Decision** answers: *“What should I do?”*
 - Input: current state, set of possible actions $\mathcal{A}$
 - Output: action $a^* \in \mathcal{A}$ that maximizes expected utility
 - Requires reasoning about **what would change** under each action
- **Question**: If we already have a highly accurate predictor, isn't the decision obvious?
 - **No**, a good predictor tells us what correlates with good outcomes, not what *causes* them

Prediction Pipeline: Standard Supervised Learning

- Typical ML system is optimized *end-to-end* for prediction:
 1. Collect observational data (X_i, Y_i)
 2. Fit a model $\hat{f}(X) \approx \mathbb{E}[Y|X]$
 3. Deploy to score new units
- **Crucial assumption:** the joint distribution $\Pr(X, Y)$ at deployment is the same as at training
 - No intervention changes how X is generated
 - No agent *acts* on the prediction
- **Example:** a spam filter trained on past emails to predict $\Pr(\text{spam}|X)$
 - Accurate predictor $\rightarrow$ useful as long as spammers do not adapt

Decision Pipeline: What Changes

- A **decision pipeline** explicitly represents:
 - The set of available actions $\mathcal{A}$
 - A utility function $U(y, a)$ over outcomes and actions
 - A **causal** model $\Pr(Y|\text{do}(a))$ of how actions affect outcomes
- **Key distinction**: $\Pr(Y|X = x)$ is observational; $\Pr(Y|\text{do}(X = x))$ is interventional
 - These can differ drastically when confounders are present
 - Decisions depend on the **interventional** quantity
- **Example**: a hospital wants to decide *whether to prescribe drug A*
 - Observational: $\Pr(\text{recovery}|\text{took drug A})$
 - Interventional: $\Pr(\text{recovery}|\text{do}(\text{give drug A}))$
 - The first is biased by who chose to take the drug

Prediction vs. Decision: Side-by-Side

Prediction

- Target: $\mathbb{E}[Y|X]$
- Data: observational only
- Metric: accuracy, AUC, MSE
- Assumes i.i.d. and stable world
- Loss if wrong: miscalibration
- No feedback with the world

- **Key takeaway:** prediction and decision-making live in different mathematical settings; collapsing them is a common source of costly errors

Decision

- Target: $\arg \max_a \mathbb{E}[U(Y) | \text{do}(a)]$
- Data: obs. + experimental
- Metric: regret, expected utility
- Models the intervention itself
- Loss if wrong: policy harm
- Actions *change* the world

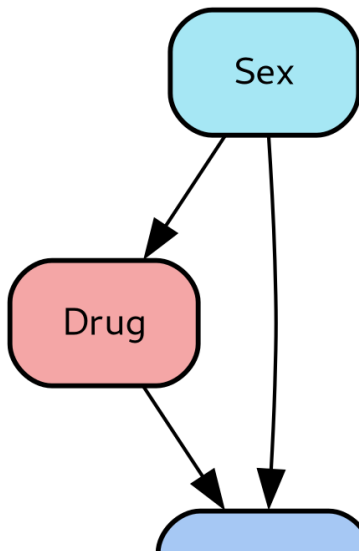
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Simpson's Paradox

- **Simpson's paradox**: a trend in subgroups reverses when groups are aggregated
 - Cause: a **confounder** is unevenly distributed across groups
 - Naive aggregation $\rightarrow$ wrong decision
- **Example**: drug recovery rates | **Group** | **Drug** (recovery) | **No Drug** (recovery) | |
————— | ————— | ————— | | Men | 81 / 87 = 93% | 234 / 270 = 87% | |
Women | 192 / 263 = 73% | 55 / 80 = 69% | | **Combined** | 273 / 350 = 78% | 289 / 350 = 83% |
- Within each subgroup, the drug helps; pooled, it appears to hurt
 - Sex is a confounder correlated with both treatment assignment and recovery baseline
 - The right decision depends on what we *do* with the drug, not what we *see*

Simpson's Paradox: Causal Resolution

- The numbers alone cannot tell us whether to prescribe the drug — we need a **causal graph**
- **Case A:** Sex $\rightarrow$ Drug, Sex $\rightarrow$ Recovery
 - Sex confounds treatment choice
 - Correct answer: look at the **subgroup** table
 - The drug helps
- **Case B:** Drug $\rightarrow$ Sex (post-treatment), Sex $\rightarrow$ Recovery
 - Sex is a mediator
 - Correct answer: look at the **aggregate** table
 - The drug hurts
- *Same data, opposite decisions* depending on the causal graph



Confounding: The Root Cause

- A **confounder** Z is a common cause of both the proposed action X and the outcome Y
 - Induces a non-causal association $X \leftrightarrow Y$ through Z
 - Observational $\Pr(Y|X) \neq \Pr(Y|\text{do}(X))$
- Without adjusting for Z , any purely predictive model will encode the spurious association
- **Example:** ice cream sales and drowning deaths are highly correlated
 - Predictor: *"more ice cream $\rightarrow$ predict more drownings"* (accurate)
 - Decision: *"ban ice cream to reduce drownings"* (disastrous)
 - Confounder: hot weather causes both
- **Q:** Can we avoid this by collecting more features?
 - Only if the missing confounder is among them *and* properly adjusted for

Policy Reversal: When Predictions Misguide Action

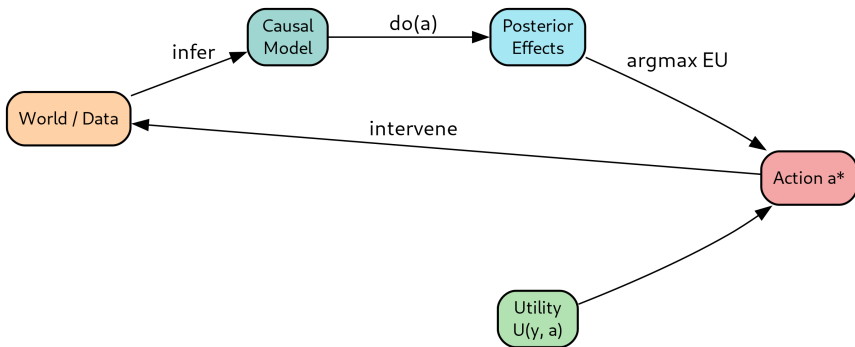
- A **policy reversal** is when an optimal-looking decision based on predictions becomes harmful once enacted
 - Common in medicine, economics, operations research
- **Mechanism**: acting on X breaks the correlation with the unobserved confounder Z , so the observed association no longer holds
- **Example**: emergency room triage
 - Predictor: patients with asthma + pneumonia have *lower* mortality
 - Naive policy: send them home first (they're low-risk)
 - Reality: they had lower mortality *because* they received aggressive care
 - Removing the care removes the protection
- **Lesson**: predictive patterns in historical data reflect the **policy that generated the data**, not what happens when we change it

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Causal Models for Decisions

- A **causal model** specifies, for each variable X_i :
 - Its direct causes (parents) $\text{Pa}(X_i)$
 - A structural mechanism $X_i = f_i(\text{Pa}(X_i), U_i)$ with noise U_i
- Enables reasoning about:
 - **Observation**: $\Pr(Y|X = x)$ (seeing x)
 - **Intervention**: $\Pr(Y|\text{do}(X = x))$ (setting x)
 - **Counterfactual**: $\Pr(Y_{X=x}|X = x', Y = y')$ (had x been different)
- For decision-making we primarily need the **interventional** layer
- **Example**: a marketing team can observe that users who open an email buy more; only a causal model can say whether *sending* the email causes purchases

Decision Loop with Causal Models



- **Key insight:** the causal model sits between data and decisions
 - Pure prediction skips the middle step
 - New data from interventions updates the model, closing the loop

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Utility Functions

- A **utility function** $U : \mathcal{Y} \rightarrow \mathbb{R}$ maps each outcome y to a real-valued preference score
 - Higher is better
 - Encodes the decision-maker's goals on a single scale
- **Properties:**
 - Utilities are *ordinal*: rank is what matters, not absolute values
 - Utilities are *subjective*: different stakeholders can have different U
 - Defined up to positive affine transformations $U \rightarrow aU + b$, $a > 0$
- **Example:** a hospital utility
 - Full recovery: $U = 1.0$
 - Recovery with complications: $U = 0.7$
 - Death: $U = 0.0$
 - A 10% complication reduction is worth 0.03 in utility on average

Expected Utility Principle

- Under uncertainty, a rational agent chooses the action a that maximizes **expected utility**:

$$a^* = \arg \max_{a \in \mathcal{A}} \mathbb{E}_{Y \sim \Pr(\cdot|a)}[U(Y)]$$

- **Von Neumann-Morgenstern theorem (1944)**: if preferences over lotteries satisfy 4 axioms (completeness, transitivity, continuity, independence), there exists a utility function U such that the agent acts *as if* maximizing expected utility
- **Example**: buying insurance
 - Premium c for certain
 - Uninsured loss L with probability p
 - Insure if $U(W - c) > p \cdot U(W - L) + (1 - p) \cdot U(W)$
 - Risk-averse utilities make this inequality easier to satisfy

Expected Utility: A Simple Example

- **Setup:** a doctor chooses between two treatments
 - Treatment 1: cures with probability 0.9, no side effect
 - Treatment 2: cures with probability 0.95, mild side effect ($U = 0.8$)
- **Utility:**
 - Cure with no side effect: 1.0
 - Cure with side effect: 0.8
 - No cure: 0.0
- **Computation:**

$$\mathbb{E}[U | T_1] = 0.9 \cdot 1.0 + 0.1 \cdot 0.0 = 0.90$$

$$\mathbb{E}[U | T_2] = 0.95 \cdot 0.8 + 0.05 \cdot 0.0 = 0.76$$

- **Decision:** choose T_1
 - Even though T_2 is more effective, its side-effect penalty outweighs the gain

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From Observational to Interventional Expectation

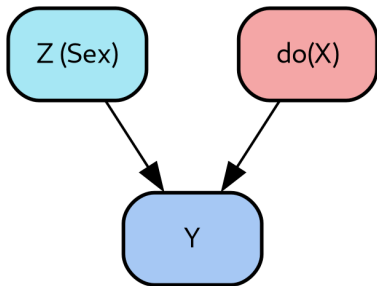
- Classical decision theory uses $\Pr(Y|a)$ — treated as the distribution of Y given action a
 - If the data is observational, $\Pr(Y|a)$ includes confounding bias
 - Plugging it into expected utility yields a **biased** action ranking
- Causal decision theory replaces it with the **interventional distribution**:

$$a^* = \arg \max_{a \in \mathcal{A}} \mathbb{E}_{Y \sim \Pr(Y|\text{do}(a))} [U(Y)]$$

- **Equivalent under**:
 - Randomized experiments (no confounders)
 - Successful back-door or front-door adjustment
 - Complete knowledge of the causal graph + sufficient data

Intervention in an Influence Graph

- **Do-operator** $\text{do}(X = x)$ mutates the causal graph:
 - Remove all edges *into* X
 - Fix X to the value x
 - Propagate through the remaining structural equations
- Turns a *passive* question into an *active* one
- **Example:** $\text{do}(\text{Drug} = 1)$
 - Drug is no longer chosen by the physician
 - All causal arrows from Sex, Age, etc. *into* Drug are cut
 - Resulting distribution reflects a policy, not the observed world



Edge $Z \rightarrow X$ is cut under $\text{do}(X)$

Expected Outcome of an Intervention

- Given a causal graph and estimated mechanisms, compute the expected outcome under each candidate action:

$$\mathbb{E}[Y | \text{do}(a)] = \sum_y y \cdot \Pr(Y = y | \text{do}(A = a))$$

- **Adjustment formula** (back-door):

$$\Pr(Y | \text{do}(A)) = \sum_Z \Pr(Y | A, Z) \Pr(Z)$$

where Z is a back-door adjustment set

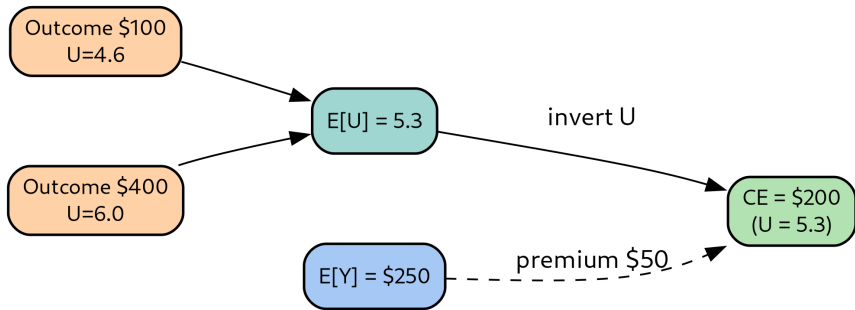
- **Example:** drug decision
 - Observational: $\Pr(\text{recovery} | \text{drug}) = 0.78$
 - Adjusted: $\sum_z \Pr(\text{rec} | \text{drug}, z) \Pr(z) = 0.83$
 - The adjusted value is what plugs into expected utility

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Risk Preferences

- Expected utility handles risk through the *shape* of U :
 - **Risk-neutral**: U is linear, $U(y) = y$ — care only about the mean
 - **Risk-averse**: U is concave — penalize variance
 - **Risk-seeking**: U is convex — reward variance
- **Certainty equivalent** y_{CE} : the sure outcome the agent values as much as a risky prospect
 - Risk-averse: $y_{CE} < \mathbb{E}[Y]$
 - Gap $\mathbb{E}[Y] - y_{CE}$ is the **risk premium**
- **Example**: $U(y) = \log(y)$
 - A 50/50 bet between \$100 and \$400 has mean \$250
 - $y_{CE} = \exp(0.5 \log 100 + 0.5 \log 400) = \$200 < \$250$

Visualizing Risk Aversion



- **Key takeaway:** a concave U trades expected return for reduced variance
 - Choose actions whose outcome distribution is *both* high in mean and narrow in spread

Multi-Criteria Trade-offs

- Real decisions weigh **multiple objectives** that pull in different directions
 - Clinical: efficacy, side effects, cost, patient burden
 - Business: revenue, churn, brand, regulatory risk
- **Scalarization**: collapse into a single utility

$$U(y) = \sum_k w_k \cdot u_k(y_k)$$

- Weights w_k elicited from stakeholders
 - Works when trade-offs are approximately linear
- **Pareto frontier**: keep only non-dominated actions
 - Avoid picking a single U until preferences are clear
 - Present trade-offs to the decision-maker visually
- **Example**: recommending a cancer therapy
 - Objectives: 5-year survival, quality of life, cost
 - Elicit weights; or present the Pareto front of treatments

Risk and Trade-offs: Practical Advice

- **Pros**
 - Expected utility unifies risk, uncertainty, and multi-criteria reasoning
 - Calibrated utilities make implicit trade-offs explicit
 - Maps cleanly onto Bayesian posteriors
- **Cons**
 - Utility elicitation is hard — stakeholders struggle to quantify preferences
 - Sensitivity: small changes in U can flip the optimal action
 - Behavioral humans *systematically* violate the axioms (Kahneman & Tversky)
- **Rule of thumb**: perform a **sensitivity analysis** over U — if the optimal action is stable across a plausible range of utilities, the decision is robust

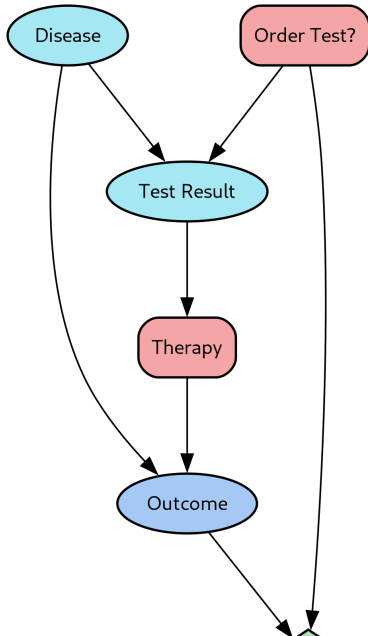
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Influence Diagrams: Motivation

- Bayesian networks model *uncertainty*; we need to also represent *decisions* and *preferences*
- An **influence diagram** extends a causal DAG with three kinds of nodes:
 - **Chance nodes** (ovals): random variables, same as in a Bayesian network
 - **Decision nodes** (rectangles): variables set by the decision-maker
 - **Utility nodes** (diamonds): deterministic functions yielding utility
- **Purpose**: a compact, graphical specification of a decision problem that can be solved by message passing
- **Example**: whether to run a diagnostic test before treating a patient
 - Chance: disease status, test result, outcome
 - Decision: order test?, choose therapy?
 - Utility: cost of test, value of outcome

Influence Diagram: Anatomy



Solving an Influence Diagram

- **Input:** influence diagram with CPTs for chance nodes and a utility table
- **Output:** optimal policy $\pi^* =$ function from information to actions
- **Steps** (variable elimination):
 1. Eliminate chance nodes by summing them out
 2. Eliminate decision nodes by **maximizing** over their domain
 3. Process in reverse topological order
 4. Record the arg-max at each decision node as the optimal policy
- **Complexity:**
 - Time: $O(|\mathcal{A}|^d \cdot \prod_i |X_i|)$ in the worst case
 - Tractable when the graph has low treewidth
- **Key insight:** the same graphical machinery used for Bayesian inference also supports optimal decision-making

Influence Diagrams: Pros and Cons

- **Pros**
 - Explicit separation of beliefs, choices, and preferences
 - Easy to communicate to non-technical stakeholders
 - Supports value of information analysis out of the box
- **Cons**
 - Requires fully specified CPTs and utilities — often heroic
 - Exact solution is intractable for large continuous problems
 - Assumes a single rational decision-maker with stable utilities
- **Example:** the oil wildcatter problem (classic)
 - Decide whether to drill, after optionally running a seismic test
 - Closed-form optimal policy using a 4-node influence diagram

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Posteriors to Optimal Actions

- Bayesian decision-making composes two steps:
 1. **Inference**: compute the posterior $\Pr(\theta|\mathcal{D})$ from data
 2. **Decision**: choose the action that maximizes *posterior* expected utility

$$a^* = \arg \max_{a \in \mathcal{A}} \mathbb{E}_{\theta \sim \Pr(\theta|\mathcal{D})} [\mathbb{E}_{Y \sim \Pr(Y|\text{do}(a), \theta)} [U(Y)]]$$

- The **inner expectation** is over outcome uncertainty given θ (aleatoric)
- The **outer expectation** is over parameter uncertainty (epistemic)
- **Key property**: Bayesian decision rule is *admissible* — no other rule dominates it under all priors

Bayesian Decision Rule: Example

- **Scenario:** A/B test of a new checkout flow
 - Parameter θ : conversion rate lift of variant B over A
 - Prior: $\theta \sim \mathcal{N}(0, 0.02^2)$
 - Data $\mathcal{D}$: $n = 5000$ users per arm; observed lift 0.01
- **Posterior:** $\theta|\mathcal{D} \sim \mathcal{N}(0.008, 0.004^2)$
 - $\Pr(\theta > 0|\mathcal{D}) \approx 0.977$
- **Utility of launching** $U(\theta) = \theta - c$, where $c = 0.002$ represents the cost of maintaining the new code
- **Expected utility:**
 - $\mathbb{E}[U|\text{launch}] = 0.008 - 0.002 = 0.006 > 0$
 - $\mathbb{E}[U|\text{keep A}] = 0$
 - **Decision:** launch B

Bayesian vs. Frequentist Decisions

Bayesian

- Posterior $\Pr(\theta|\mathcal{D})$
- Acts on full distribution
- Incorporates prior beliefs
- Natural for sequential updating
- Connects directly to utility
- **Pros**
 - Coherent under uncertainty
 - Works for small samples
- **Cons**
 - Requires priors
 - Computationally heavier
- **Bottom line:** Bayesian framing slots naturally into causal decision-making because both use joint distributions end-to-end

Frequentist

- Point estimate $\hat{\theta} + \text{CI}$
- Acts on test outcomes
- No priors
- Decision via hypothesis test
- Utility often implicit
- **Pros**
 - Fewer subjective inputs
 - Standardized in many fields
- **Cons**
 - Thresholds (e.g., $p < 0.05$) are not utility-optimal
 - Harder to combine with new data

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Prior Elicitation: What and Why

- In causal decision problems, the prior encodes beliefs about the **causal effect** θ before seeing data
 - Size, sign, and uncertainty of θ
- Prior elicitation is especially important when:
 - Data is scarce or expensive (medicine, policy)
 - Experiments are unethical or impossible
 - Stakeholders disagree
- **Q**: Where do priors come from?
 - Literature meta-analyses
 - Expert judgment
 - Past experiments or pilot studies
 - Regulatory or legal constraints

Methods for Eliciting Priors

- **Quantile elicitation**: ask experts for 10/50/90 percentiles of θ , then fit a distribution
 - E.g., “10% chance lift is below 0.5%, 50% below 1%, 90% below 3%”
- **Pairwise judgments**: elicit relative probabilities of outcomes, combine via a log-linear model
- **Historical calibration**: compare past predictions against actuals to adjust for bias and overconfidence
- **Reference priors**: use weakly informative priors (Jeffreys, unit information) when knowledge is vague
 - Avoid flat priors in high dimensions — they are not uninformative
- **Example**: a clinical trial team estimates a 20% prior probability that a new drug is at least as effective as standard of care, based on meta-analysis of 14 similar compounds

Pitfalls and Best Practices

- **Pitfalls**
 - **Anchoring**: experts cling to a first-mentioned number
 - **Overconfidence**: intervals are consistently too narrow
 - **Framing effects**: elicit the *same* quantity two ways — answers differ
 - **Confusing** aleatoric vs. epistemic uncertainty — see later section
- **Best practices**
 - Elicit from multiple experts independently, then aggregate
 - Cross-validate with subsets of historical data
 - Perform **prior sensitivity analysis**: does the decision change under plausible alternative priors?
 - Document the elicitation process — decisions may need defending

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Value of Information: Motivation

- Before committing to a costly action, a decision-maker may gather more data
 - Diagnostic test, survey, pilot experiment, additional A/B arm
- **Q**: Is the extra information worth its cost?
- **Value of information (VOI)** quantifies the *expected* gain in utility from obtaining new evidence *before* making a decision
 - A central concept in decision analysis
- **Example**: a company considers an expensive market survey before a \$10M product launch
 - Launch without: expected utility $\mathbb{E}[U]$
 - Launch with survey outcome E : $\mathbb{E}_E[\max_a \mathbb{E}[U(Y) | \text{do}(a), E]]$
 - Run the survey only if the difference exceeds its cost

Expected Value of Perfect Information (EVPI)

- **EVPI** is the upper bound on what any information-gathering activity could be worth:

$$\text{EVPI} = \mathbb{E}_{\theta} [\max_a U(a, \theta)] - \max_a \mathbb{E}_{\theta} [U(a, \theta)]$$

- Interpret the two terms:
 - First: utility if we could learn θ *perfectly* before deciding
 - Second: utility of the best action under current uncertainty
- **Key properties:**
 - $\text{EVPI} \geq 0$ always
 - $\text{EVPI} = 0$ iff the current best action is optimal for *every* θ
 - Any test costing more than EVPI is not worth running, under any accuracy

Expected Value of Sample Information (EVSI)

- Real information sources are **imperfect**
 - EVSI relaxes EVPI to consider noisy evidence E
 - $EVSI(E) = \mathbb{E}_E[\max_a \mathbb{E}[U|a, E]] - \max_a \mathbb{E}[U|a]$
- $0 \leq EVSI(E) \leq EVPI$
 - Noisier evidence $\rightarrow$ smaller EVSI
 - Tighter prior $\rightarrow$ smaller EVSI (less to learn)
- **Example:** diagnostic test with sensitivity 0.9 and specificity 0.85
 - Compute EVSI by integrating over all possible ($E = +, E = -$) outcomes
 - Run the test if $EVSI - \text{cost} > 0$
- **Practical tip:** run a quick EVPI screen first — if EVPI is tiny, skip the EVSI computation entirely

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Exploration vs. Exploitation

- A sequential decision-maker faces a tension:
 - **Exploit**: pick the action that *currently looks best*
 - **Explore**: try other actions to *reduce uncertainty* about their effects
- Pure exploitation can lock in a suboptimal choice based on early noise
 - Pure exploration wastes samples on clearly bad options
- **Example**: a doctor who always prescribes the first drug that worked early in her career will never learn about newer, better options
- Causal models add richness: exploration can target *structural* uncertainty, not just parameter uncertainty

Causal Multi-Armed Bandits

- Classic multi-armed bandit: K arms with unknown reward distributions
 - At each step, pull one arm, observe reward, update beliefs
 - Objective: minimize **regret** relative to the best arm
- **Causal bandit**: arms are *interventions* $\text{do}(a)$ on a known (or partly known) causal graph
 - Observing one arm provides information about *other* arms via the graph
 - Much lower regret than a standard bandit in many settings
- **Example**: advertising
 - Arms: combinations of channel, audience, creative
 - Causal structure: audience and creative *interact*, channel has additive effect
 - Exploiting this structure cuts samples needed to find the best combination

Thompson Sampling

- **Thompson sampling:** a Bayesian exploration strategy
 - Maintain a posterior over each arm's mean reward
 - At each step, sample a parameter from each posterior and pull the arm with the highest sampled mean
- **Algorithm:**
 1. Start with prior $\Pr(\theta_a)$ for each arm a
 2. Sample $\tilde{\theta}_a \sim \Pr(\theta_a|\mathcal{D})$ for each a
 3. Pull $a^* = \arg \max_a \tilde{\theta}_a$
 4. Observe reward, update posterior, repeat
- **Properties:**
 - Naturally balances exploration and exploitation
 - Near-optimal regret bounds $O(\sqrt{TK \log T})$
 - Easy to adapt to causal structure by sampling from posteriors over causal effects

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Bayesian Optimization: The Setup

- **Bayesian optimization (BO)** targets problems where:
 - The objective $f(x)$ is **expensive** to evaluate (one experiment per evaluation)
 - No gradients available
 - x lies in a low- to moderate-dimensional space
- **Recipe:**
 1. Place a prior over f , typically a **Gaussian process**
 2. At each round, fit the GP posterior on observed $(x_i, f(x_i))$
 3. Select x_{next} maximizing an **acquisition function**
 4. Query $f(x_{next})$, update, repeat
- **Example use cases:**
 - Hyperparameter tuning for a deep model
 - Tuning web page layout
 - Picking reagent concentrations in a chemistry lab

Acquisition Functions

- The acquisition function trades off exploration and exploitation:
 - **Expected improvement (EI)**: $\alpha_{EI}(x) = \mathbb{E}[\max(f(x) - f^*, 0)]$
 - **Upper confidence bound (UCB)**: $\alpha_{UCB}(x) = \mu(x) + \kappa\sigma(x)$
 - **Thompson sampling**: sample $\tilde{f} \sim \Pr(f|\mathcal{D})$, maximize $\tilde{f}$
 - **Entropy search**: pick x that most reduces entropy of $\arg \max f$
- **Example**: tuning a learning rate
 - Posterior mean peaks at 10^{-3} , variance is largest at 10^{-5}
 - EI may query 10^{-5} if improvement potential $>$ current best
 - UCB with high κ behaves similarly
- **Key takeaway**: the acquisition function *is* the policy, derived from the posterior and a utility for information

BO in Causal Experimentation

- Causal BO combines BO with causal graphs:
 - Decision variable x is an **intervention** $\text{do}(X = x)$
 - Surrogate model is a **structural causal model**, not just a GP on y
 - Leverages the graph to share information across interventions
- **Example**: drug dose optimization
 - Standard BO models $y = f(\text{dose})$ directly
 - Causal BO models intermediate biomarkers that mediate dose $\rightarrow$ outcome
 - Fewer trials needed when the causal mechanism is partly known
- **Pros**: sample efficient, principled exploration, uncertainty-aware
- **Cons**: requires accurate causal structure; misspecification can hurt more than help

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Aleatoric Uncertainty: Irreducible Randomness

- **Aleatoric uncertainty** comes from the *inherent* randomness in the outcome
 - Also called *statistical* or *risk* uncertainty
 - Cannot be reduced by collecting more data
 - Captured by $\Pr(Y | \text{do}(a), \theta)$ given known θ
- **Example**: flipping a known-fair coin
 - $\Pr(\text{heads}) = 0.5$ exactly
 - More flips don't sharpen the single-flip outcome
- **Example**: medical treatment with a known response distribution
 - Even if we knew the drug's exact mechanism, individual patient outcomes would vary due to biological stochasticity
- **Decision implication**: aleatoric uncertainty sets the *floor* on risk — no amount of learning removes it

Representing Aleatoric Uncertainty

- In a structural causal model:

$$Y = f(\text{Pa}(Y), U_Y), \quad U_Y \sim \Pr(U_Y)$$

- U_Y is the exogenous noise capturing aleatoric variability
- Common distributional assumptions:
 - Gaussian: $U_Y \sim \mathcal{N}(0, \sigma^2)$ for continuous Y
 - Bernoulli / categorical for discrete outcomes
 - Heavy-tailed (Student- t , Laplace) for robust modeling
- **Q**: How do we tell the agent apart from the noise when we only see Y ?
 - We can't, unless U_Y has a tractable parametric form and we exploit the causal structure
- **Example**: a recommender's click-through rate varies user-to-user; even with perfect knowledge of $\Pr(\text{click}|\text{user}, \text{item})$, specific clicks are random

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Epistemic Uncertainty: Model Misspecification

- **Epistemic uncertainty** stems from *lack of knowledge*
 - Also called *systematic* or *knowledge* uncertainty
 - Reducible in principle by collecting more/better data or improving the model
 - Captured by the posterior $\Pr(\theta|\mathcal{D})$
- **Sources:**
 - Limited or biased data
 - Wrong functional form
 - Missing variables or wrong causal graph
 - Confounders not adjusted for
- **Example:** a small clinical trial with $n = 40$ patients
 - Wide posterior on treatment effect θ
 - Gathering more data shrinks this posterior

Aleatoric vs. Epistemic: Side-by-Side

Aleatoric

- Source: inherent randomness
- Reducible: **No**
- Represented: noise U
- Grows with: complexity of Y
- Managed by: risk preferences
- Example: coin flip, weather
- **Implication**: influences utility function shape

- **Key takeaway**: good decision systems *separate* the two — combining them collapses information that matters for action

Epistemic

- Source: model & data limits
- Reducible: **Yes**, via evidence
- Represented: posterior on θ
- Shrinks with: data, experiments
- Managed by: learning, VOI
- Example: effect of a new drug
- **Implication**: motivates sequential decision-making

Distinguishing in Practice

- A Bayesian predictive distribution decomposes as:

$$\Pr(Y | \text{do}(a), \mathcal{D}) = \int \Pr(Y | \text{do}(a), \theta) \Pr(\theta | \mathcal{D}) d\theta$$

- **Variance decomposition:**

$$\mathbb{V}[Y | \text{do}(a), \mathcal{D}] = \underbrace{\mathbb{E}_{\theta}[\mathbb{V}[Y | a, \theta]]}_{\text{aleatoric}} + \underbrace{\mathbb{V}_{\theta}[\mathbb{E}[Y | a, \theta]]}_{\text{epistemic}}$$

- **Example:** deep ensemble for sales forecasting
 - Within-model variance $\approx$ aleatoric
 - Between-model variance $\approx$ epistemic
 - Report both to the decision-maker
- **Warning:** most off-the-shelf ML models report only a point prediction — neither kind of uncertainty is visible

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Communicating Uncertainty to Stakeholders

- Stakeholders often prefer a single number; giving one is a disservice
- **Best practices:**
 - Report **intervals** (90% posterior credible intervals) alongside point estimates
 - Distinguish aleatoric from epistemic in words and visuals
 - Frame probabilities in natural frequencies: *“about 2 in 100”*
 - Use **visual aids**: fan charts, violin plots, calibration plots
 - Link uncertainty to **actions**: *“if we get 200 more samples, the interval shrinks from \$5M to \$1M”*
- **Anti-patterns:**
 - Reporting only point predictions
 - Combining aleatoric and epistemic into one opaque bar
 - Hiding priors in appendices

Communicating Uncertainty: A Worked Example

- **Setting:** pricing team evaluating a new fee structure
- **Bad communication:**
 - *“The model predicts \$2.3M in additional revenue.”*
- **Better communication:**
 - *“Our best estimate is \$2.3M. 90% of the uncertainty sits between \$0.8M and \$3.7M.*
 - *Of this, about \$1.5M of the width is intrinsic customer variability (aleatoric) — a pilot won’t narrow that much.*
 - *The remaining \$1.4M is parameter uncertainty (epistemic) and would shrink by roughly 60% if we ran the proposed 4-week pilot.”*
- **Decision impact:** the pilot discussion now has a clear expected value, not a gut feeling

Summary: Decision-Making with Causal Models

- **Prediction alone is insufficient** for choosing actions — Simpson's paradox, confounding, and policy reversal all require causal reasoning
- **Expected utility** under interventional distributions is the right target

$$a^* = \arg \max_a \mathbb{E}_{Y \sim \Pr(Y | \text{do}(a))} [U(Y)]$$

- **Influence diagrams** extend causal DAGs with decisions and utilities, giving a graphical, solvable decision problem
- **Sequential decisions** need exploration: VOI tells us when to gather information; Thompson sampling and Bayesian optimization tell us *how*
- **Uncertainty** decomposes into aleatoric (irreducible) and epistemic (reducible) — both must be modeled and communicated
- **Key takeaway**: causal reasoning, decision theory, and Bayesian inference are three sides of the same coin — and together they turn predictions into *robust* actions